Polynomial End Behavior & Zeros

ANSWER KEY

Section 1: Leading Term Test and End Behavior

1 Determine the end behavior of the polynomial function $f(x) = -3x^5 + 7x^3 - 2x + 9$.

a. **As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$** ✓

b. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$

c. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$

d. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$

The leading term is $-3x^5$. Because the degree 5 is odd and the leading coefficient -3 is negative, $f(x) \to -\infty$ as $x \to \infty$, and $f(x) \to \infty$ as $x \to -\infty$.

2 If a polynomial $p(x)$ satisfies $\lim_{x \to \infty} p(x) = -\infty$ and $\lim_{x \to -\infty} p(x) = -\infty$, which of the following could be the leading term of $p(x)$?

a. $-4x^6$ ✓

b. $2x^6$

c. $-5x^7$

d. $3x^5$

When both ends approach $-\infty$, the polynomial must have an even degree and a negative leading coefficient. Among the choices, $-4x^6$ has even degree 6 and negative coefficient -4.

3 Consider $g(x) = (2 - x)(x + 3)^2(2x - 1)$. What is the leading coefficient of $g(x)$?

a. -2 ✓

b. 2

c. -1

d. 4

Expanding the highest power terms: $(-x) \cdot (x)^2 \cdot (2x) = -2x^4$. Thus, the leading coefficient is -2.

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Section 2: Multiplicity of Zeros and Graph Geometry

4 Let $f(x) = (x-4)^3(x+2)^2(x-1)$. At which zero does the graph of f touch the x-axis and turn around without crossing?

a. $x = -2$ ✓

b. $x = 4$

c. $x = 1$

d. $x = 2$

A graph touches the x-axis and turns around at zeros with even multiplicity. Here, $(x+2)^2$ has multiplicity 2 (even), so $x = -2$ is a turning touch point. At $x = 4$ and $x = 1$, the graph crosses.

5 A polynomial $p(x)$ of degree 4 has a zero at $x = 3$ of multiplicity 2, and no other real zeros. If $p(0) = 18$, which equation represents $p(x)$ if its remaining zeros are non-real complex conjugates $\pm i$?

a. $p(x) = 2(x-3)^2(x^2+1)$ ✓

b. $p(x) = (x-3)^2(x^2+1)$

c. $p(x) = 2(x-3)^2(x^2-1)$

d. $p(x) = \frac{1}{2}(x-3)^2(x^2+1)$

Zeros at $\pm i$ correspond to factor (x^2+1). So $p(x) = a(x-3)^2(x^2+1)$. Using $p(0) = a(-3)^2(1) = 9a = 18 \Rightarrow a = 2$. Thus $p(x) = 2(x-3)^2(x^2+1)$.

6 At $x = c$, $f(x) = (x-c)^3 g(x)$ where $g(c) \neq 0$. What is the behavior of the graph of f at $(c, 0)$?

a. **The graph crosses the x-axis with an inflection/flattening point (horizontal tangent)** ✓

b. The graph touches the x-axis and turns around symmetrically

c. The graph has a vertical asymptote at $x = c$

d. The graph crosses the x-axis linearly without flattening

An odd multiplicity greater than 1 ($m = 3$) causes the graph to cross the x-axis while flattening out (creating an inflection point with a horizontal tangent).

Section 3: Remainder & Factor Theorems

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7 When $f(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x - 3)$, the remainder is 7. Find the value of k.

a. -4 ✓

b. 4

c. -2

d. 5

By the Remainder Theorem, $f(3) = 7$. $f(3) = 2(27) - 5(9) + 3k - 8 = 54 - 45 + 3k - 8 = 1 + 3k$. Setting $1 + 3k = 7 \Rightarrow 3k = 6 \Rightarrow k = 2$.

8 Which of the following is a factor of $P(x) = x^3 - 6x^2 + 11x - 6$?

a. $x - 2$ ✓

b. $x + 2$

c. $x - 4$

d. $x + 1$

Evaluate $P(2) = 2^3 - 6(4) + 11(2) - 6 = 8 - 24 + 22 - 6 = 0$. Since $P(2) = 0$, by the Factor Theorem $(x - 2)$ is a factor.

9 A polynomial $q(x)$ satisfies $q(-4) = 0$, $q(1) = 0$, and $q(5) = 12$. Which statement must be true?

a. $(x + 4)$ **and** $(x - 1)$ **are factors of** $q(x)$ ✓

b. $(x - 4)$ and $(x + 1)$ are factors of $q(x)$

c. The remainder when $q(x)$ is divided by $(x + 5)$ is 12

d. $q(x)$ must be a quadratic polynomial

By the Factor Theorem, $q(c) = 0 \Leftrightarrow (x - c)$ is a factor. Here $q(-4) = 0 \Rightarrow (x + 4)$ is a factor, and $q(1) = 0 \Rightarrow (x - 1)$ is a factor.

Section 4: Rational Root Theorem & Complex Conjugates

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10 According to the Rational Root Theorem, which of the following is NOT a possible rational zero of $f(x) = 3x^3 - 2x^2 + 5x - 4$?

a. $\frac{3}{2}$ ✓

b. $\frac{4}{3}$

c. ± 1

d. ± 2

Factors of constant term -4: $\pm 1, \pm 2, \pm 4$. Factors of leading coefficient 3: $\pm 1, \pm 3$. Possible rational roots p/q are $\pm 1, \pm 2, \pm 4, \pm 1/3, \pm 2/3, \pm 4/3$. Thus $\frac{3}{2}$ is not possible.

11 A polynomial with real coefficients has degree 3. Two of its zeros are 4 and $2 - 3i$. What is the third zero?

a. $2 + 3i$ ✓

b. $-2 + 3i$

c. -4

d. $-2 - 3i$

By the Complex Conjugate Root Theorem, non-real complex zeros of polynomials with real coefficients must occur in conjugate pairs. Since $2 - 3i$ is a zero, $2 + 3i$ must also be a zero.

12 What is the total number of complex zeros (counting multiplicity) of $P(x) = (x^2 + 4)^2(x - 5)^3$?

a. 7 ✓

b. 5

c. 4

d. 3

The degree of $P(x)$ is $2 \times 2 + 3 = 7$. By the Fundamental Theorem of Algebra, a polynomial of degree n has exactly n complex zeros counting multiplicity.

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